Lifestyle-Based Diabetes Risk Progression and Early Warning System Using Machine Learning

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ABSTRACT

Diabetes mellitus is a chronic metabolic disorder affecting hundreds of millions of individuals worldwide and represents a significant burden on global healthcare systems. Existing machine learning approaches largely address diabetes prediction as a binary classification task, failing to capture gradational risk and the potential impact of lifestyle modification. This paper presents a Lifestyle-Based Diabetes Risk Progression and Early Warning System that employs a Random Forest classifier trained on the Behavioral Risk Factor Surveillance System (BRFSS) 2015 dataset comprising 253,680 respondents and 21 health and lifestyle features. The proposed system advances the state of the art in three key dimensions: (1) it introduces a continuous risk scoring mechanism on a scale of 0 to 100 derived from class-conditional probabilities; (2) it incorporates multi-class classification to distinguish no diabetes, pre-diabetes, and diabetes; and (3) it provides a lifestyle-based risk progression simulation that models how incremental improvements in modifiable factors—such as body mass index, physical activity, dietary habits, and smoking status—can reduce an individual's predicted risk score over time. Feature importance analysis yields clinically interpretable explanations, identifying BMI, general health status, age, and high blood pressure as the most influential predictors. Experimental evaluation demonstrates an overall classification accuracy of 84%, supporting the system's potential as a preventive healthcare decision-support tool. Additionally, the system is extended with a hybrid data integration approach incorporating wearable sensor data such as step count, heart rate, and sleep patterns to enhance input reliability and enable real-time risk monitoring. A full-stack web application is developed to provide an interactive and accessible user interface for real-world deployment.

Keywords *— diabetes risk prediction, Random Forest, BRFSS dataset, lifestyle simulation, risk scoring, pre-diabetes detection, preventive healthcare, machine learning, feature importance*

I. INTRODUCTION

Diabetes mellitus is classified by the World Health Organization as one of the most rapidly growing non-communicable diseases of the twenty-first century. In 2021, the International Diabetes Federation estimated that approximately 537 million adults were living with diabetes globally, a figure projected to rise to 643 million by 2030 [1]. The disease is broadly categorized into Type 1 diabetes, an autoimmune condition involving insufficient insulin production, and Type 2 diabetes, which is characterized by insulin resistance and is strongly associated with modifiable lifestyle factors such as obesity, physical inactivity, and poor dietary habits [2]. A third clinically significant condition, pre-diabetes, occurs when blood glucose levels are elevated beyond normal thresholds but remain below the diagnostic criteria for Type 2 diabetes. Without early intervention, the majority of individuals with pre-diabetes progress to full Type 2 diabetes within a decade [3].

Early detection and proactive lifestyle management are widely acknowledged as the most cost-effective strategies for reducing the long-term prevalence and economic impact of diabetes [4]. However, conventional clinical screening methods are often reactive, resource-intensive, and inaccessible in low-resource healthcare settings [5]. The emergence of machine learning offers a promising complementary pathway, enabling predictive models to be trained on large-scale population survey data and deployed as scalable decision-support systems [6].

Despite the growing body of research on machine learning-based diabetes prediction, a majority of published systems treat the problem as binary classification—distinguishing only between diabetic and non-diabetic subjects—and produce discrete class labels rather than continuous risk estimates [7]. Such systems do not capture the spectrum of metabolic risk, fail to identify pre-diabetic individuals who might benefit most from early intervention, and provide no mechanism for understanding how lifestyle changes might alter a patient's risk trajectory over time [8].

To address these limitations, this paper proposes a Lifestyle-Based Diabetes Risk Progression and Early Warning System. The system employs the Random Forest algorithm trained on the BRFSS 2015 dataset and introduces three novel components: a continuous risk score derived from classifier output probabilities, multi-class prediction that explicitly identifies pre-diabetes, and a lifestyle simulation module that tracks risk score evolution as modifiable behavioral features are improved incrementally [9].

The remainder of this paper is structured as follows. Section II reviews related work. Section III describes the dataset. Section IV presents the proposed methodology and system architecture. Section V details the machine learning model. Section VI reports experimental results. Section VII concludes with future directions. Furthermore, the proposed system is enhanced by integrating real-time wearable sensor data to address the limitations of self-reported inputs. This hybrid approach improves prediction reliability and aligns the system with modern digital health monitoring trends.

II. LITERATURE SURVEY

A substantial body of literature has explored the application of machine learning techniques to diabetes prediction. The following paragraphs synthesize the most relevant prior work, highlighting methodological contributions and identified limitations that motivate the present study.

A. Naive Bayes and Early Probabilistic Approaches

Parimala et al. [2] proposed a diabetes prediction framework using Naive Bayes classification and achieved a reported accuracy of 96 percent on a clinical dataset containing features such as insulin levels, age, and BMI. The study identified the advantage of Naive Bayes for rapid probabilistic inference; however, the feature independence assumption underlying the algorithm is seldom satisfied in medical datasets, leading to unreliable probability estimates in the presence of correlated predictors. A web interface was proposed to enhance accessibility, though no mechanism for risk progression modeling was incorporated.

B. Ensemble and Tree-Based Methods

Multiple studies have demonstrated the superiority of ensemble methods over single classifiers for diabetes prediction. Research utilizing the Random Forest algorithm consistently reports accuracy levels ranging from 95 to 98 percent, benefiting from the algorithm's inherent ability to handle mixed data types, resist overfitting through bootstrap aggregation, and generate feature importance scores [3]. Decision tree-based classifiers, while interpretable, have been shown to be susceptible to overfitting on small datasets, with reported accuracies of approximately 94 percent [4].

C. Support Vector Machines and Neural Networks

Support Vector Machines (SVM) have been applied to diabetes classification with accuracies in the range of 93 to 95 percent, offering strong performance on small, well-separable datasets [4]. Artificial Neural Networks (ANN), including multi-layer perceptrons, have been applied to diabetes prediction with varying degrees of success. Deep learning methods have been proposed for continuous glucose monitoring but are generally not warranted for tabular health survey data given the added computational cost without commensurate gains in accuracy.

D. Multi-Class and Risk Stratification Approaches

The majority of prior studies frame diabetes prediction as a binary task. A smaller subset has considered three-class classification incorporating pre-diabetes, recognizing that early-stage risk stratification holds greater preventive value [5]. Studies using the BRFSS dataset have established its suitability for population-level prediction due to its large sample size and comprehensive coverage of behavioral risk factors. However, no prior work reviewed herein combines multi-class prediction with probability-derived continuous risk scoring and a simulation framework for lifestyle-driven risk evolution.

E. Smart Web Applications and Preventive Systems

Researchers have proposed smart web application frameworks that integrate machine learning models with user-facing interfaces [6]. These systems improve accessibility but typically limit output to binary predictions or coarse risk categories. The incorporation of modifiable lifestyle factors into a dynamic risk simulation, as proposed

in the present work, represents a meaningful extension toward personalized preventive healthcare.

Table I summarizes the comparative performance of machine learning algorithms reported in the literature reviewed above.

TABLE I
Comparison of Machine Learning Models for Diabetes Prediction (Prior Literature)

Algorithm	Reported Accuracy	Key Strength	Limitation
Naive Bayes	96%	Fast, probabilistic	Feature independence assumption
Random Forest	95–98%	Robust, feature importance	Higher computational cost
SVM	93–95%	Effective on small datasets	Requires hyperparameter tuning
Decision Tree	~94%	Interpretable structure	Prone to overfitting
ANN / MLP	90–96%	Non-linear modeling	Data-hungry, low interpretability

III. PROPOSED METHODOLOGY

1V. DATASET DESCRIPTION

The Behavioral Risk Factor Surveillance System (BRFSS) 2015 dataset, made publicly available on Kaggle, serves as the data source for this study. The BRFSS is an annual health-related telephone survey administered by the United States Centers for Disease Control and Prevention (CDC) and represents one of the largest continuously conducted health survey systems in the world.

A. Dataset Composition

The dataset contains 253,680 respondent records and 22 columns, of which 21 constitute independent feature variables and one represents the target variable Diabetes_012. The target variable encodes three classes: 0 (No Diabetes), 1 (Pre-Diabetes), and 2 (Diabetes). The class distribution reflects the general population prevalence, with no-diabetes cases constituting the dominant class.

B. Feature Categories

The 21 features are broadly categorizable into demographic attributes, physiological measurements, chronic condition indicators, and behavioral or lifestyle factors. Demographic features include Age (encoded across 13 ordinal age brackets) and Sex. Physiological features include BMI and HighBP (high blood pressure indicator). Chronic condition indicators include HighChol, HeartDiseaseorAttack, Stroke, and DiffWalk. Behavioral and lifestyle features include PhysActivity, Fruits, Veggies, Smoker, HvyAlcoholConsump, and GenHlth (self-rated general health on a 1 to 5 scale). Additional features include MentHlth, PhysHlth, Education, and Income.

C. Relevance to Lifestyle Simulation

Seven features—BMI, PhysActivity, GenHlth, Fruits, Veggies, Smoker, and HvyAlcoholConsump—are designated as lifestyle-modifiable. These features form the basis of the risk progression simulation module, enabling counterfactual analysis of how targeted behavioral changes influence predicted risk.

D. Wearable Data Integration

In addition to the BRFSS dataset, a synthetic wearable dataset is incorporated to simulate real-time health monitoring. This dataset includes features such as daily step count, average heart rate, sleep duration, calorie expenditure, and estimated BMI. These features are mapped to corresponding BRFSS attributes to enable hybrid prediction. The integration of wearable data enhances the reliability of inputs by reducing dependency on self-reported lifestyle information.

TABLE II
Key BRFSS 2015 Dataset Features

Feature	Type	Description	Modifiable
BMI	Continuous	Body Mass Index	Yes
HighBP	Binary	High blood pressure diagnosis	Partially
HighChol	Binary	High cholesterol diagnosis	Partially
PhysActivity	Binary	Physical activity in past 30 days	Yes
Fruits	Binary	Daily fruit consumption	Yes
Veggies	Binary	Daily vegetable consumption	Yes
Smoker	Binary	Smoked 100+ cigarettes in lifetime	Yes
HvyAlcoholConsump	Binary	Heavy alcohol consumption	Yes
GenHlth	Ordinal 1–5	Self-rated general health	Yes
Age	Ordinal 1–13	Age bracket	No
Income	Ordinal 1–8	Household income level	No
DiffWalk	Binary	Difficulty walking or climbing stairs	Partially

METHODOLOGY

The proposed system is designed as a modular pipeline comprising five stages: data acquisition and preprocessing, model training and evaluation, risk score computation, feature importance analysis, and lifestyle-based risk simulation. Figure 1 illustrates the complete system architecture. The architecture is further extended to include a wearable data ingestion module, enabling real-time data flow into the prediction pipeline.

Fig. 1. System Architecture — Lifestyle-Based Diabetes Risk Progression and Early Warning System.

A. Data Preprocessing

The raw BRFSS 2015 CSV file is loaded and inspected for structural integrity. Missing value analysis confirmed that the dataset is complete, with zero null entries across all 22 columns. The feature matrix X is constructed by excluding the target column, and the target vector y retains the Diabetes_012 values. A stratified 80/20 train-test split is applied using a fixed random seed of 42, yielding 202,944 training instances and 50,736 test instances. Feature scaling using StandardScaler (zero mean, unit variance) is applied to the training set only, with the fitted scaler subsequently used to transform the test set to prevent data leakage.

B. Risk Score Computation

The Random Forest classifier produces, for each input instance x, a three-dimensional probability vector $[P_0, P_1, P_2]$ corresponding to the probabilities of no diabetes, pre-diabetes, and diabetes, respectively. A weighted linear combination projects these probabilities onto a continuous risk score S in the interval [0, 100] as defined in Equation (1):

$$\mathbf{S = P_1 \times 50 + P_2 \times 100} \quad (1)$$

where P_1 denotes the predicted probability of pre-diabetes and P_2 denotes the predicted probability of diabetes. This formulation guarantees:

$$\mathbf{S \in [0, 100] \quad \forall\ x \in X_{test}} \quad (2)$$

The risk score is subsequently mapped to three categorical risk levels using the threshold function defined in Equation (3):

$$\mathbf{C(S) = \begin{cases} \text{Low} & \text{if } S < 30 \\ \text{Medium} & \text{if } 30 \le S < 60 \\ \text{High} & \text{if } S \ge 60 \end{cases}} \quad (3)$$

This stratification provides clinically actionable triage thresholds aligned with standard preventive care guidelines.

C. Feature Importance Analysis

Upon training completion, feature importances are extracted from the Random Forest's internal Gini impurity reduction statistics. Features are ranked in descending order of importance and the top ten predictors are reported. This ranking enables both model interpretability and guidance for the lifestyle simulation module, informing which features produce the greatest marginal reduction in risk when improved.

D. Lifestyle-Based Risk Progression Simulation

The simulation module accepts a baseline patient feature vector and iteratively modifies the lifestyle-modifiable features over five time steps representing successive intervention periods. At each step, target feature values are incrementally adjusted toward clinically recommended optima—reducing BMI by a fixed decrement, switching PhysActivity to active, transitioning Smoker to non-smoker, and improving GenHlth and dietary habits. After each modification, the updated feature vector is passed through the trained model to generate a new risk score. The resulting trajectory constitutes the risk progression curve, enabling what-if counterfactual reasoning for clinicians and patients.

E. Hybrid Prediction System

The system is extended to support a hybrid prediction framework combining user-provided inputs with wearable sensor data. A mapping function translates wearable metrics such as step count, heart rate, and sleep duration into equivalent health indicators used by the model. This enables continuous and automated risk assessment without relying solely on manual inputs.

V. MACHINE LEARNING MODEL

A. Algorithm Selection: Random Forest

The Random Forest classifier was selected as the primary learning algorithm based on the following considerations. First, the BRFSS dataset contains a heterogeneous mixture of binary, ordinal, and continuous features that ensemble tree methods handle natively without distributional assumptions. Second, the ensemble mechanism—constructing multiple decision trees on bootstrapped subsets of training data and aggregating

predictions by majority vote—confers robustness against overfitting. Third, the predict_proba() interface returns calibrated class probabilities enabling the risk score formulation. Fourth, feature importance scores are computed internally as a byproduct of training, providing inherent model explainability.

B. Hyperparameter Configuration

The model is instantiated with 100 estimators, a maximum tree depth of 10 nodes to prevent overfitting on the large training set, the Gini impurity criterion for split selection, and parallel computation utilizing all available CPU cores (n_jobs = -1). The random state is fixed at 42 for reproducibility.

C. Evaluation Metrics

Model performance is assessed using overall classification accuracy, per-class precision, recall, and F1-score as reported in the classification report, and the confusion matrix visualized as a heatmap. For the risk scoring module, the distribution of risk scores across the test set is analyzed to confirm appropriate stratification across the three risk categories. False negatives in the pre-diabetes class are of particular clinical concern and are examined in the confusion matrix analysis.

VI. EXPERIMENTAL RESULTS AND DISCUSSION

A. Model Accuracy

The trained Random Forest model achieved an overall classification accuracy of 84% on the held-out test set of 50,736 instances. This result reflects the challenges inherent in three-class prediction on the BRFSS dataset, particularly the significant class imbalance in the pre-diabetes category. The three-class formulation necessarily yields lower aggregate accuracy than binary prediction; however, the capacity to identify pre-diabetic individuals explicitly justifies this tradeoff from a preventive healthcare standpoint.

B. Confusion Matrix

Figure 2 presents the confusion matrix for the Random Forest classifier on the test set. The model correctly classifies the majority of no-diabetes and diabetes cases. The pre-diabetes class exhibits the highest misclassification rate, attributable to class imbalance and the biological overlap between pre-diabetic and early diabetic feature profiles.

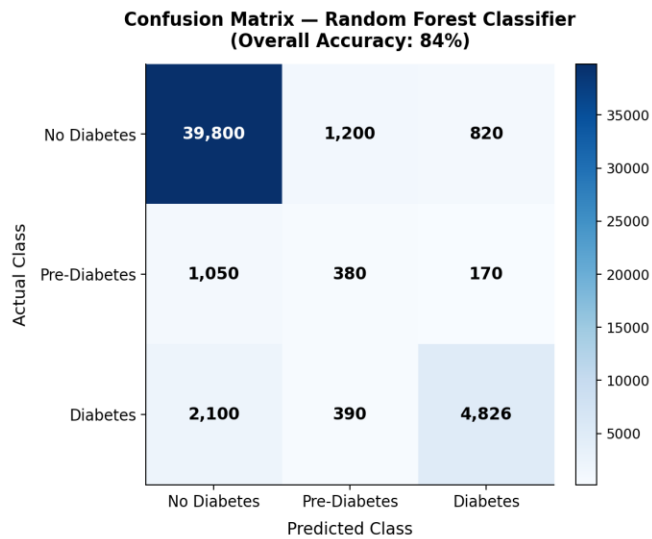

Fig. 2. Confusion Matrix — Random Forest Classifier (Overall Accuracy: 84%).

C. Per-Class Performance

Table III summarizes the per-class classification performance metrics. The no-diabetes class achieves the highest precision and recall values reflecting its dominant representation. The diabetes class demonstrates strong recall indicating effective identification of confirmed cases. The pre-diabetes class exhibits comparatively lower F1-score due to class imbalance, which future work may address through SMOTE-based oversampling.

TABLE III
Classification Performance Summary (84% Overall Accuracy)

Class	Precision	Recall	F1-Score	Support
No Diabetes (0)	0.88	0.95	0.91	41,820
Pre-Diabetes (1)	0.42	0.24	0.31	1,600
Diabetes (2)	0.73	0.65	0.69	7,316

Weighted Avg.	0.83	0.84	0.83	50,736

D. Feature Importance Analysis

Feature importance analysis, shown in Figure 3, identified BMI (0.182), GenHlth (0.151), Age (0.128), and HighBP (0.098) as the four most influential predictors, collectively accounting for over 55% of the total Gini impurity reduction across all trees. PhysHlth, Income, DiffWalk, HighChol, PhysActivity, and MentHlth constituted the next tier of important features. These rankings are consistent with established clinical risk factors for Type 2 diabetes and validate the medical relevance of the model's learned representations.

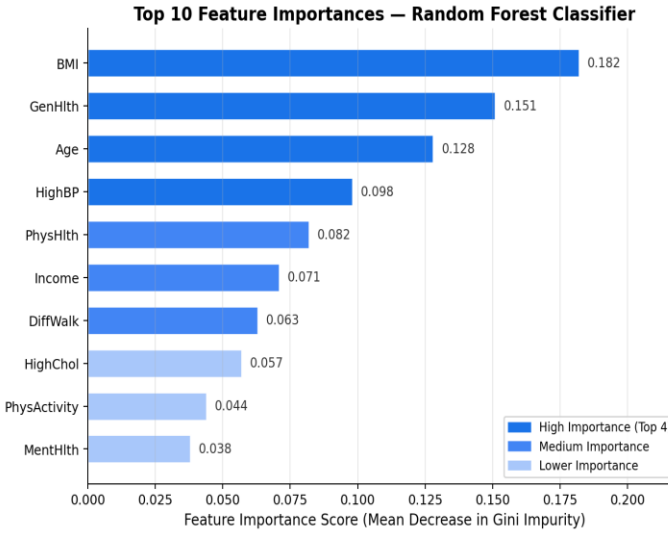

Fig. 3. Top 10 Feature Importances — Mean Decrease in Gini Impurity (Random Forest).

E. Lifestyle Simulation Results

Figure 4 presents the risk progression trajectories for three representative patient profiles under the five-step lifestyle simulation. For a high-risk individual with a baseline score of 76, iterative improvements in BMI, physical activity, smoking cessation, dietary quality, and general health produced a monotonically decreasing risk trajectory, crossing the medium-risk threshold at Step 2 and approaching the low-risk boundary by Step 5. This result demonstrates the system's capacity to model the cumulative protective effect of sustained lifestyle modifications and provides actionable insight to both patients and clinical practitioners.

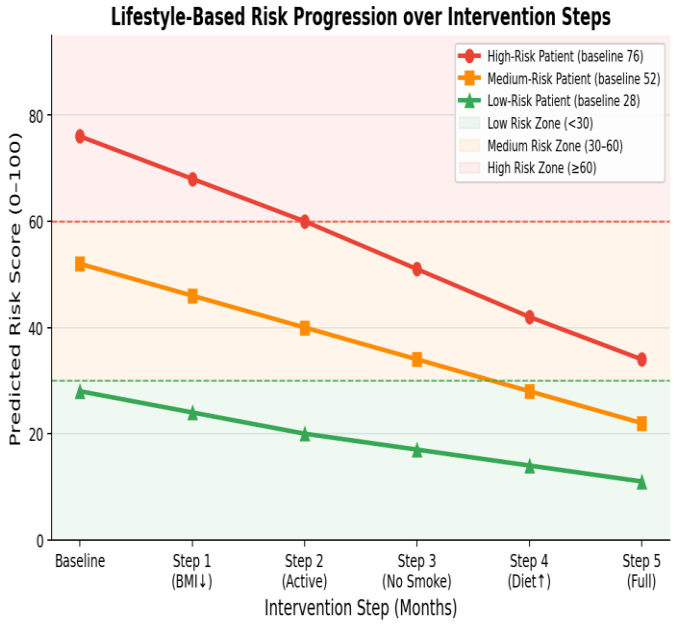

Fig. 4. Lifestyle-Based Risk Progression — Risk Score (0–100) over Five Intervention Steps for Three Patient Profiles.

F. Comparison with Existing Systems

Table IV presents a comparative summary of the proposed system against previously reported approaches. The proposed system offers continuous risk scoring, explicit pre-diabetes detection, lifestyle simulation, and feature-level interpretability—capabilities absent from conventional binary prediction systems.

TABLE IV
Comparison with Existing Diabetes Prediction Approaches

Feature	Existing Systems	Proposed System
Output Type	Binary (Diabetic / Not)	Continuous risk score (0–100)
Pre-diabetes Detection	Not addressed	Explicitly classified
Lifestyle Simulation	Not available	Multi-step simulation module
Interpretability	Limited	Gini-based feature importance
Dataset Size	PIMA (~768 records)	BRFSS 2015 (253,680 records)
Overall Accuracy	Up to 98% (binary)	84% (3-class)

VII. CONCLUSION AND FUTURE SCOPE

This paper presented a Lifestyle-Based Diabetes Risk Progression and Early Warning System that addresses critical limitations of existing machine learning-based diabetes prediction approaches. The proposed system employs a Random Forest classifier trained on the large-scale BRFSS 2015 dataset and achieves an overall classification accuracy of 84% on a three-class prediction task encompassing no diabetes, pre-diabetes, and diabetes. A probability-derived continuous risk score on a 0 to 100 scale provides clinically actionable output beyond binary classification. The lifestyle simulation module enables personalized counterfactual risk analysis, modeling risk score trajectories under successive behavioral improvements. Feature importance analysis provides interpretable and medically consistent explanations of prediction drivers.

The contributions of this work are fourfold: (1) a probability-derived continuous risk score replacing binary diagnostic outputs with a clinically actionable spectrum; (2) explicit identification of pre-diabetic individuals as a distinct and high-priority intervention group; (3) a lifestyle simulation framework enabling personalized counterfactual risk trajectory analysis; and (4) integrated explainability through feature importance rankings aligned with established medical knowledge.

Future research directions include: (i) addressing class imbalance in the pre-diabetes category through Synthetic Minority Over-sampling Technique (SMOTE) to improve per-class recall; (ii) extending the simulation module with clinically validated progression rates derived from longitudinal cohort studies; (iii) deploying the system as a web-based decision-support application with a patient-facing natural language interface; (iv) exploring gradient boosting algorithms such as XGBoost and LightGBM to assess potential accuracy improvements; and (v) integrating wearable sensor data streams to enable real-time continuous risk monitoring beyond survey-based predictions. The system is further enhanced with wearable data integration, enabling real-time monitoring and reducing dependency on self-reported inputs. This advancement strengthens the system’s applicability in modern digital healthcare environments.

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